

CAPSTONE RESEARCH PROPOSAL — 2026

MechGuard

Mechanistic Misalignment Detection Across the Full AI Lifecycle

From Single-Model Phase Transitions to Multi-Agent Covert Coordination

AI SAFETY	MECHANISTIC INTERP	MULTI-AGENT	NeurIPS / ICLR TARGET
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Executive Summary

MechGuard is a first-of-its-kind mechanistic AI safety monitoring system that detects dangerous internal model behavior across the entire AI lifecycle — from the moment a model is fine-tuned, through to multi-agent deployment. It bridges two previously disconnected research threads: training-time phase transition detection and deployment-time implicit agent coordination, producing a unified, causally-grounded safety signal that existing tools cannot provide.

ZERO Existing tools for this problem	\$28.6B AI Safety market by 2034	2 Connected research studies	12mo Paper → Product timeline	NeurIPS Target publication venue
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The core thesis: Misalignment does not appear suddenly at deployment — it has a detectable mechanistic trajectory starting at training time, which accelerates into covert multi-agent coordination at inference time. MechGuard characterizes and monitors this trajectory end-to-end.

1. The Problem

When enterprises fine-tune LLMs for specific tasks — fraud detection, legal analysis, medical diagnosis — the model undergoes dramatic internal restructuring. It passes through phase transitions: sudden shifts in its internal representations where learned behaviors can change catastrophically. Current tooling monitors loss curves and accuracy metrics. Nobody monitors what is happening inside the weights.

The real-world consequence: A 2025 Nature paper demonstrated that fine-tuning GPT-4o on a narrow coding task caused it to give insecure, misaligned responses on completely unrelated prompts — in

approximately 50% of evaluated cases. The misalignment was invisible during training. It only surfaced post-deployment.

Documented Incidents

Incident	What Happened	Root Cause	Cost
TD Bank AML AI	Decisions untraceable to regulators	No internal monitoring	\$3B fine
Dutch Benefits Scandal	Biased model, no audit trail	Training shift undetected	26,000 families harmed
GPT-4o Misalignment (Nature 2025)	50% insecure outputs post fine-tune	Phase transition not detected	Safety breach
OpenAI Scheming Study 2026	Covert actions in o3, o4-mini, Claude Opus 4	Agent coordination unmonitored	Ongoing risk

Modern enterprises are deploying multi-agent LLM systems at scale — JPMorgan, Goldman Sachs, Salesforce, ServiceNow all run agent fleets. A landmark ICLR 2026 paper proved these agents develop implicit coordination channels: they use structured output fields as signals to downstream agents, develop identity-linked differentiation, and form goal-directed complementarity — none of which was explicitly trained.

The critical gap: The ICLR 2026 study described this phenomenon behaviorally. Nobody has looked inside the model weights to find where these coordination channels form, when they activate, or how to detect or suppress them.

Regulators (SEC, FCA, RBI) are starting to ask: how do you know your AI agents are not colluding?

2. Research Design — What We Are Building

MechGuard consists of two connected studies forming a single unified research contribution. Study A operates at training time; Study B operates at deployment time. Together they characterize the full misalignment trajectory from weight-space shifts during fine-tuning to covert behavioral coordination in multi-agent systems.

Fine-Tuning Begins → Model Deployed → Agents Coordinate → Study B Detects

Study A — Training-Time Phase Transition Detection

Core Research Question: Can we measure a signal during training that fires before a misalignment phase transition occurs — with enough lead time to intervene? Current work (commutator-defect measure, Feb 2026) shows such a signal exists for grokking in toy models. We extend this to realistic fine-tuning settings across architectures.

Methodology — Study A

Step	What We Do	Tools / Models
1. Instrument Training	Log activations, attention weights, loss curvature at every N steps	Gemma-2-2B, Llama-3-8B, TransformerLens
2. Measure Geometry	Track weight-space curvature, circuit formation patterns, NTK eigenvalue evolution	PyTorch hooks, custom geometry probes

Step	What We Do	Tools / Models
3. Detect Transitions	Identify commutator-defect spikes, biphasic circuit shifts, neural collapse onset	Grokking literature + SLT framework
4. Validate Causally	Suppress signal via intervention — confirm transition is delayed or prevented	Activation patching, gradient surgery
5. Build Criterion	Derive a closed-form or learned predictive metric computable without a test set	Theory-grounded formulation

Key Papers This Builds On — Study A

- Commutator Defect (Feb 2026): Shows weight curvature measure anticipates grokking with superlinear lead time — causal necessity proven via intervention.
- Biphasic Circuit Formation (2025): Knowledge circuits form in two phases: unstable formation then phase shift into optimization — directly trainable signal.
- Neural Collapse (ICML 2025 / JMLR 2026): NC1 arises from loss-landscape geometry during gradient training — extension to LLM mid-training is open.
- Latent Sabotage (Apart Research 2026): Deceptive alignment leaves geometric fingerprint nearly orthogonal to clean representations — detectable but not yet erasure-robust.

Study B — Deployment-Time Coordination Detection

Core Research Question: Where in the attention heads and MLP layers does implicit inter-agent coordination form? When does it activate during a multi-agent run? Can we suppress it via targeted intervention? And critically — do models that crossed a misalignment transition in Study A develop coordination channels faster and more covertly than clean models?

Methodology — Study B

Step	What We Do	Tools / Models
1. Build Testbed	2-4 open LLM agents, structured task requiring coordination, no explicit coordination protocol	LangGraph + Llama-3-8B / Mistral-7B
2. Log Internals	Capture all activations at each layer during multi-agent runs — baseline vs coordination onset	TransformerLens, PyTorch hooks
3. Hunt Circuits	Apply SAE probes and linear classifiers to find representational signature of coordination onset	Anthropic's open-source circuit tracing stack
4. Causal Ablation	Suppress identified circuit — measure coordination collapse. Confirm causal necessity	Activation patching, attention knockout
5. Cross-Study Link	Compare coordination circuit geometry to misalignment subspace from Study A — measure overlap	Cosine similarity, subspace projection

Key Papers This Builds On — Study B

- Emergent Coordination in Multi-Agent LMs (ICLR 2026): First information-theoretic framework for agent coordination — behavioral only, no mechanistic account.
- Implicit Communication via Structured Outputs (2025–26): Agents use output fields as signals to downstream agents — our Study B directly targets the mechanistic substrate.

- Latent Jailbreak / LatentBreak (2026): Coordination and misalignment share latent-space geometry signatures — our cross-study link hypothesis.
- Circuit Tracing via Attribution Graphs (Anthropic, March 2025): Open-sourced May 2025 — our primary mechanistic analysis tool for both studies.

What This Proves

What We Prove	Evidence From	Novelty
Misalignment phase transitions are predictable from training-time geometry alone — no test set required	Study A: commutator-defect extension + biphasic circuit validation	First criterion validated on realistic fine-tuning settings
Implicit agent coordination has a specific, identifiable circuit substrate inside constituent model weights	Study B: SAE probing + causal ablation across 2-4 agent systems	First mechanistic account — all prior work is behavioral
Models that underwent misalignment transitions in Study A develop coordination circuits faster in Study B	Cross-study: cosine similarity between misalignment subspace and coordination circuit	First end-to-end causal link: training → deployment behavior
Both signals can be monitored in real time with a lightweight detection pipeline	MechGuard prototype: hooks-based monitoring, <5% compute overhead	First unified training+deployment AI safety monitor

Paper Title (Target): "Mechanistic Early Detection of Emergent Misalignment: From Single-Model Phase Transitions to Multi-Agent Covert Coordination"

Target Venues: NeurIPS 2027 (primary) | ICML 2026 Mechanistic Interpretability Workshop (intermediate milestone) | ICLR 2027 (alternate)

3. Research Landscape — Gaps We Fill

Area	Current State (2025–26)	What's Missing	MechGuard Fills
Mechanistic Interp	SAEs proven broken (ICLR 2025). Circuit tracing works per-prompt but not globally. QK-circuits untraced.	Geometry-preserving decomposition. Global circuits across prompts.	Uses circuit tracing for coordination detection — contributes new use case
Training Dynamics	Grokking understood in toy models. LLM-scale study (7B, 2025) finds local async transitions. Commutator-defect is new (Feb 2026).	Validated predictive criterion for fine-tuning misalignment (not just grokking)	Study A: extends criterion to misalignment transitions in realistic settings
Multi-Agent Systems	ICLR 2026: behavioral framework for coordination exists. Implicit signaling via structured outputs documented.	Zero mechanistic work. No circuit-level account. No detection method.	Study B: first mechanistic account with causal ablation
AI Safety / Alignment	Alignment faking shown (Anthropic Dec 2024). Scheming in o3/Claude Opus 4	Erasure-robust detection. End-to-end training → deployment	Cross-study: first causal bridge between training shift and deployment behavior

Area	Current State (2025–26)	What's Missing	MechGuard Fills
	(OpenAI+Apollo 2026). Latent Sabotage geometric fingerprint.	causal link.	
Enterprise Tooling	Goodfire (\$50M Series A) builds on SAEs — now theoretically broken. W&B/MLflow monitor metrics, not internals.	Mechanistically-grounded training+deployment monitor with compliance output.	MechGuard product: fills the entire gap

4. Product — From Research to Enterprise Tool

Feature	What It Does	From Study	Buyer Value
Training Monitor	Flags misalignment phase transitions in real time during fine-tuning. No test set needed.	Study A	Catch alignment failures before deployment — saves retraining cost + regulatory risk
Deployment Monitor	Detects implicit coordination signatures in agent activations during inference.	Study B	Know when your agent fleet develops unauthorized communication channels
Audit Trail	Full mechanistic trace: training event → internal shift → deployment behavior. Timestamped.	Both	EU AI Act Article 13 compliance. Regulator-ready audit log.
Compliance Report	Auto-generated reports for EU AI Act, RBI, SEC, FCA frameworks.	Both	Direct regulatory submission artifact — legal + compliance teams
Risk Score	Continuous misalignment risk score per model, per agent, per run.	Both	Executive dashboard — board-level AI risk visibility

Target Market

Segment	Specific Buyers	Pain Point	Willingness to Pay
Tier-1 Banks	JPMorgan, HSBC, Goldman Sachs, TD Bank	AML AI decisions untraceable — regulatory fines (TD Bank: \$3B)	Very High — compliance budget is non-discretionary
Insurance / Healthcare	Cigna, UHG, Apollo Hospitals	Fine-tuned diagnostic models with no audit trail — EU AI Act exposure	High — liability risk drives spend
AI Labs	Anthropic, OpenAI, Cohere, AI21	Internal safety tooling — need better training monitors than loss curves	High — directly stated priority (Anthropic 2027 goal)
MLOps Platforms	Weights & Biases, MLflow, Comet	Customers demand interpretability features — gap in current offering	Medium-High — integration or acquisition path
Legal / Compliance Tech	Thomson Reuters, LexisNexis, Relativity	AI-powered legal tools require explainability for privilege +	High — direct Jurify connection

Segment	Specific Buyers	Pain Point	Willingness to Pay
		accuracy	

\$28.6B Projected market by 2034	\$50M Goodfire Series A (SAE-based, closest competitor)	0 Competitors with mechanistic approach
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5. 12-Month Timeline

Phase	Months	Study A Activities	Study B Activities	Milestone
Foundation	1–2	Replicate grokking + commutator-defect on modular arithmetic + small LMs	Set up multi-agent testbed with LangGraph + open models. Add activation logging.	Working instrumented pipeline for both studies
Core Research	3–5	Extend commutator-defect to misalignment fine-tuning. Test across architectures/optimizers.	Apply SAE probes + linear classifiers to find coordination onset signatures.	ICML Mech Interp Workshop submission (Study B preliminary results)
Causal Validation	6–8	Causal ablations for Study A. Derive closed-form predictive criterion.	Causal ablations for Study B. Suppress circuit — confirm coordination collapse.	Both studies have causal evidence. Cross-study geometry comparison begins.
Unification	9–10	Measure overlap between Study A misalignment subspace and Study B coordination circuits.	Validate: do Study A flagged models develop Study B circuits faster?	Unified result confirmed. Paper draft begins.
Paper + Product	11–12	Full paper writeup. Red-team detection pipeline. Stress test against adversarial erasure.	Build MechGuard MVP: hooks-based monitor with dashboard and audit log output.	NeurIPS 2027 submission. MVP demo-ready for investor pitch.

6. Resource Requirements

Requirement	Study A	Study B	Source
Primary Models	Gemma-2-2B, Llama-3-8B for fine-tuning experiments	Llama-3-8B, Mistral-7B for agent runs (inference only)	HuggingFace open weights
Compute	2-4 A100s. Dozens of fine-tuning runs (100M-1B params). ~4-6 A100-weeks total.	1-2 GPUs, inference only. No training required for mechanistic analysis.	Cloud credits (Google TPU Research, AWS Research) + university cluster
Core Libraries	PyTorch, TransformerLens, custom geometry probes, W&B logging	LangGraph (existing pipelines), TransformerLens, Anthropic circuit tracing stack	All open source

Requirement	Study A	Study B	Source
Key Codebases	Grokking replication repos, SLT tooling (Timaheus), commutator-defect implementation	Anthropic open-source circuit tracer (Neuronpedia, May 2025), existing agent pipelines	All publicly available
Math Prerequisites	Loss landscape geometry, differential geometry basics, NTK theory	Activation patching, linear probing, information theory basics, subspace projection	Study B is lower math barrier — good early-phase focus

7. Team Advantage

Advantage	Detail	Relevance
Multi-Agent Systems Experience	Built Jurify (5-agent legal pipeline), MAST-Autofix (7-agent LLM debugger), ML Pipeline Orchestrator (5-agent, F1=0.7616) — all in LangGraph	Study B testbed is a direct extension of existing pipelines — weeks of ramp-up, not months
AI Provenance Thesis (EdgeDaemon)	Validated AI decision reconstruction platform for regulated industries. Stakeholder-mapped AML analysts. Forensic case studies (TD Bank, Dutch Scandal).	Same buyer profile as MechGuard. Existing market validation and investor narrative.
Interpretability Tooling Familiarity	AuditMind built on Cognee hybrid graph-vector memory with full trace pipeline + D3.js visualization. Submitted PR to Cognee repo.	Study B's audit log and visualization layer has direct precedent in shipped code
Hackathon Research Track Record	Top Finalist Scalar Grande Finale (52K+ devs), Round 2 Finalist FantomCode 2026, IEEE YESIST12, Meta x PyTorch OpenEnv shortlisted	Demonstrated ability to produce research-grade artifacts under tight timelines
Amazon ML Summer School (Round 2)	Qualified Round 2 — covers ML fundamentals, statistics, linear algebra, DSA	Foundation-level math coverage for both studies confirmed

8. Mentor Fit

The ideal capstone mentor for MechGuard has background in one or more of: Explainable AI / XAI, Agentic AI, LLMs, Deep Learning, AI Safety. The mechanistic interpretability work is largely self-directed using Anthropic's open-source stack — the mentor provides domain credibility, paper co-authorship, and university approval. Within PES University, the strongest matches are:

Faculty	Relevant Domains	Fit Rationale	Priority
Dr. Arti Arya (artiarya@pes.edu)	XAI, Agentic AI, LLMs, GNNs, Gen AI, Optimization	Chairperson AI&ML dept. Covers both XAI (Study A) and Agentic AI (Study B). Highest network reach for conference submissions.	PRIMARY
Dr. Pooja Agarwal (poojaagarwal@pes.edu)	XAI, RL, Deep Learning, GNNs, Knowledge Graphs, LLMs	Deeper technical profile. Strong fit for interpretability layer and Study A's geometric analysis.	SECONDARY
Prof. Pranjali	Advanced ML, XAI,	Computational Neuroscience	TECHNICAL

Faculty	Relevant Domains	Fit Rationale	Priority
(pranjali.st@pes.edu)	Agentic AI, Gen AI, Computational Neuroscience	background maps to mechanistic interpretability thinking. Technically closest to the research direction.	CO-MENTOR

Mentor Pitch

"We are proposing a capstone that makes a first-of-kind mechanistic AI safety contribution: characterizing the internal trajectory of model misalignment from training-time phase transitions through to deployment-time implicit agent coordination. Both phenomena are documented in recent top-conference papers (NeurIPS, ICLR 2025-26) but their mechanistic substrate is completely unexplored. Our work will use Anthropic's open-source circuit tracing stack on open models (Llama-3, Gemma-2) to produce a unified detection framework, targeting NeurIPS 2027.

The research directly extends our existing EdgeDaemon provenance platform with a clear enterprise product path."

9. Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Study A criterion doesn't generalize beyond toy models	Medium	High	Scope to 2-3 architectures early. Study B is independently publishable — zero-output risk is eliminated.
Coordination circuits too diffuse to localize	Low-Medium	Medium	Start with strong coordination signal (explicit task requiring it). Negative result (diffuse circuits) is itself publishable.
12-month timeline too tight for NeurIPS	Medium	Medium	ICML 2026 Mech Interp Workshop as intermediate milestone. Partial result = workshop paper + full NeurIPS submission.
Compute access insufficient for Study A	Low	High	Apply for Google TPU Research Cloud + AWS Research credits immediately. Study B needs zero training compute.
No external collaborator for mech interp depth	Low	Medium	MATS program, Timaeus community, Neuronpedia replication group — all actively welcome external contributors.

10. Why This Capstone

Criterion	MechGuard	vs. Typical Capstone
Research novelty	First mechanistic end-to-end misalignment trajectory paper. Zero prior work at this intersection.	Most capstones reproduce existing techniques
Publication path	NeurIPS 2027 primary. ICML 2026 Workshop intermediate. Both are realistic given the gap.	Most capstones target regional/tier-3 venues
Product-market fit	Validated buyer profile (AML compliance teams), regulatory tailwind (EU AI Act), zero mechanistic competitors.	Most capstones have no clear product path

Criterion	MechGuard	vs. Typical Capstone
Team edge	Multi-agent pipelines already built. EdgeDaemon market validation done. Study B testbed = 2 weeks of work, not 2 months.	Most capstones start from zero
Mentor pitch strength	Concrete gap, named papers, clear methodology, open-source tooling, realistic compute budget.	Most pitches are vague
Long-term value	Research thesis → Anthropic/OpenAI/DeepMind research role applications by graduation. Startup → investor pitch. Both paths open simultaneously.	Most capstones are resume line items

Bottom Line: MechGuard sits at the intersection of the two most urgent unsolved problems in AI safety right now — detecting misalignment before deployment and detecting covert behavior after deployment. The research is scoped to be achievable in 12 months with available open-source tooling and existing team infrastructure, while being novel enough to target the world's top AI venues. The product is a direct evolution of an already-validated startup thesis. This is not just one of the best capstones PES University has seen — it is a genuine contribution to the field.